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CHAPTER 1

STRING THEORY

1.1 Partition function for a point particle and bosonic string

Partition functions play a fundamental role in theoretical physics, as they encode complete information about the state of a theory and act as generating functionals for correlation functions.

In string theory, it is crucial to distinguish between the worldsheet and the spacetime descriptions. This distinction already arises in the case of point particles. In standard quantum field theory courses, one typically develops the formalism from a spacetime perspective, where particles are described as fields propagating in spacetime. However, particles admit two equivalent descriptions: a spacetime (field-theoretic) perspective and a worldline perspective.

In what follows, we describe the partition functions of particles and strings using the worldline and worldsheet formulations, respectively, and establish their connection with the spacetime partition function.

The canonical partition function Z in quantum statistical mechanics is defined as

$$Z = \text{Tr}_{\mathcal{H}} e^{-\beta H}$$

where $\beta = (k_B T)^{-1}$ denotes the inverse temperature, k_B is the Boltzmann constant, H is the Hamiltonian, and the trace is taken over the Hilbert space $\mathcal{H}$ of the theory.

The partition function of a quantum theory can be obtained through a Wick rotation of the trace of the time-evolution operator. Starting from [1]

$$Z = \text{Tr} U(t_2, t_1) = \text{Tr} e^{-i(t_2 - t_1)H} = \sum_{\phi} \langle \phi | e^{-i(t_2 - t_1)H} | \phi \rangle$$

where the sum runs over an orthonormal basis $\{|\phi\rangle\}$, one performs the analytic continuation $t_2 - t_1 \rightarrow -i\beta$, leading to $Z = \text{Tr} e^{-\beta H}$.

The trace corresponds to a finite time evolution in which the initial and final states

are identified. Diagrammatically, the time-evolution operator $U(t_2, t_1)$ is represented by a line, and taking the trace amounts to identifying its endpoints and summing over all states, resulting in a closed loop, as shown in Fig. 1.1. The length of this loop corresponds to the proper time t required for the particle to circle around.

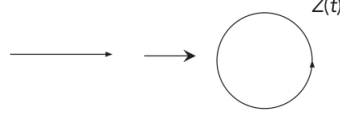


Figure 1.1: The worldline partition function $Z(t)$ is constructed from the propagator by identifying the initial and final states and summing over all possible states. It depends explicitly on the proper time t required for the particle to complete the loop.

From the spacetime perspective, one integrates over all values of the proper time t . However, one may define an unintegrated partition function $Z(t)$, known as the *worldline partition function*, which depends explicitly on the proper time[2].

We now extend this discussion to closed strings. Diagrammatically, the propagator of a closed string is represented by a cylinder rather than a line. Identifying the initial and final states of the propagator corresponds to gluing the ends of the cylinder, thereby producing a torus[2]. In contrast to the point-particle case, closed strings admit an additional possibility: the string may be twisted before the cylinder is closed. Equivalently, the initial and final states can differ by a translation along the spatial direction of the string.

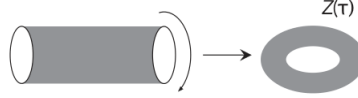


Figure 1.2: The worldsheet partition function $Z(\tau)$ of a closed string is obtained from the propagator by identifying the initial and final states and summing over all intermediate states. It depends on both the length of the cylinder and the relative shift applied when gluing the final state to the initial one, which together are encoded in the complex modulus τ .

More precisely, a torus with complex modulus $\tau = \tau_1 + i\tau_2$ can be viewed as arising from a quantum field theory defined on a circle, evolved for Euclidean time $2\pi\tau_2$ while simultaneously translated along the spatial coordinate σ^1 by an amount $2\pi\tau_1$, followed by identification of the endpoints[3]. In operator language, this construction leads to the trace

$$Z = \text{Tr}_{\mathcal{H}} \left(e^{2\pi i \tau_1 P - 2\pi \tau_2 H} \right) \quad (1.1)$$

where $H = L_0^{\text{cyl}} + \bar{L}_0^{\text{cyl}}$, $P = L_0^{\text{cyl}} - \bar{L}_0^{\text{cyl}}$ are the worldsheet Hamiltonian and momentum operators, respectively. Here L_0^{cyl} and $\bar{L}_0^{\text{cyl}}$ denote the Laurent zero modes of the energy-momentum tensor defined on the cylinder.

It is important to distinguish these operators from their counterparts L_0 and $\bar{L}_0$ on the complex plane, since the energy - momentum tensor is a quasi-primary field. Under a conformal transformation $z \rightarrow \tilde{z}(z)$, the stress tensor transforms as[4]

$$T(z) \longrightarrow \tilde{T}(\tilde{z}) = \left(\frac{d\tilde{z}}{dz} \right)^{-2} \left[T(z) - \frac{c}{12} S(z, \tilde{z}) \right]$$

where c is the central charge and $S(z, \tilde{z})$ is the Schwarzian derivative,

$$S(z, \tilde{z}) = \frac{\partial^3 \tilde{z}}{\partial z^3} \left(\frac{\partial \tilde{z}}{\partial z} \right)^{-1} - \frac{3}{2} \left(\frac{\partial^2 \tilde{z}}{\partial z^2} \right)^2 \left(\frac{\partial \tilde{z}}{\partial z} \right)^{-2}$$

For the conformal map from the cylinder to the complex plane, $z(w) = e^{\frac{2\pi}{\ell} w}$, the Schwarzian derivative evaluates to $S(z, w) = -\frac{1}{2} \left(\frac{2\pi}{\ell} \right)^2$. As a result, the stress tensor on the cylinder is related to the stress tensor on the plane by

$$T_{\text{cyl}}(w) = \left(\frac{2\pi}{\ell} \right)^2 \left[z^2 T_{\text{plane}}(z) - \frac{c}{24} \right]$$

Using this relations, we can show that

$$L_0^{\text{cyl}} = L_0 - \frac{c}{24}$$

Observe that the L_0 eigenvalues shift proportional to the central charge[2]. Introducing $q = e^{2\pi i \tau}$ and $\bar{q} = e^{-2\pi i \bar{\tau}}$, equation (1.1) can be rewritten as[2]

$$Z(\tau) = \text{Tr}_{\mathcal{H}} \left(q^{L_0^{\text{cyl}}} \bar{q}^{\bar{L}_0^{\text{cyl}}} \right) = \text{Tr}_{\mathcal{H}} \left(q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}} \right) \quad (1.2)$$

At this point, we temporarily set aside the discussion of strings and turn to the computation of particle partition functions within this formalism.

Partition function of a free point particle

The einbein (worldline metric) form of the action is

$$S = \frac{1}{2} \int d\tau (e^{-1} \dot{x}^\mu \dot{x}_\mu - e m^2)$$

where $x^\mu(t)$ is the particle worldline and $e(t)$ is the einbein.

The canonical momentum conjugate to x^μ is $p_\mu = \frac{\partial L}{\partial \dot{x}^\mu} = e^{-1} \dot{x}_\mu$. For a free particle and having fixed the gauge, the worldline Hamiltonian is given by $H = k^2 + m^2$, where k denotes the particle momentum. The trace then reduces to a Gaussian integral. Neglecting overall normalization constants, for a particle propagating in a one-dimensional target space one finds[2]

$$Z(t) = \text{Tr} e^{-tH} = \text{Tr} e^{-t(k^2 + m^2)} = \int dk e^{-t(k^2 + m^2)} \approx t^{-1/2} e^{-tm^2} \quad (1.3)$$

This expression represents the *worldline partition function* for a single circular connected worldline of fixed Euclidean proper time t .

In general, the partition function in quantum field theory includes a sum over all numbers of disconnected circular worldlines, reflecting the fact that the theory is formulated on a multiparticle Hilbert space. Taking the logarithm of the partition function removes the combinatorial contributions from disconnected loops, isolating the contribution of a single connected loop[5].

To compute the one-loop effective action¹, defined as the logarithm of the spacetime

¹Remember that the effective action, which generates all one-particle-irreducible (1PI) diagrams, admits a loop expansion. The one-loop effective action $\Gamma^{(1)}$ contains all 1PI diagrams with exactly one loop.

partition function, $\Gamma^{(1)} = -\ln Z$, one must sum over all connected circular worldlines modulo gauge symmetries. This involves integrating over the proper time t while implementing worldline reparametrization invariance and gauge fixing. The resulting integration measure $\frac{dt}{t}$ emerges as the Faddeev–Popov measure after fixing reparametrization invariance and dividing by the volume of the residual translation symmetry along the loop[6].

For a particle propagating in D -dimensional spacetime, the full partition function is obtained by integrating with measure dt/t over D copies of the one-dimensional worldline partition function[2].

We now compute the partition function for a massive scalar field from the spacetime perspective and show its relation to the worldline partition function $Z(t)$. The Euclidean action for a free massive scalar field in D dimensions is $S[\phi] = \int d^D x \phi(-\Delta + m^2)\phi$. The associated partition function is formally given by[5]

$$Z = \int \mathcal{D}\phi e^{-S[\phi]}$$

where $\mathcal{D}\phi$ indicates integration over all field configurations. Using the standard Gaussian integral formula, this can be evaluated as[5]

$$Z = \det^{-1/2}(-\Delta + m^2)$$

Taking the logarithm, we obtain the standard QFT expression for the one-loop partition function:

$$\ln Z = \ln \det^{-1/2}(-\Delta + m^2) = -\frac{1}{2} \text{tr} \ln(-\Delta + m^2)$$

where the trace is over the eigenvalues of the operator $-\Delta + m^2$. Using the momentum representation, this becomes[2]

$$\ln Z = -\frac{1}{2} V_D \int \frac{d^D k}{(2\pi)^D} \log(k^2 + m^2)$$

with V_D the spacetime volume, which we treat as a constant even if formally infinite.

To relate this to the worldline partition function, we use the Schwinger proper-time representation of the logarithm[4]:

$$-\ln(k^2 + m^2) \rightarrow \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{\infty} \frac{dt}{t} e^{-(k^2 + m^2)t}$$

Although the integral diverges as $\epsilon \rightarrow 0$, it can be regularized via its relation to the exponential integral function (see equations ?? and ??). Using this representation, the partition function becomes

$$\ln Z = \frac{1}{2} V_D \int \frac{d^D k}{(2\pi)^D} \int_0^{\infty} \frac{dt}{t} e^{-(k^2 + m^2)t}$$

The Gaussian integral over momenta can then be performed, yielding

$$\ln Z \approx V_D \int_0^{\infty} \frac{dt}{t} t^{-D/2} e^{-m^2 t} \tag{1.4}$$

up to numerical constants. Comparing this with the fixed-time worldline partition function

$Z(t)$, we find[2]

$$\ln Z \approx V_D \int_0^\infty \frac{dt}{t} Z^D(t)$$

As expected, the effective one-loop action in D -dimensional spacetime can be written as an integral over D copies of the worldline partition function for a one-dimensional target space, with the correct gauge-invariant measure. The integration over t corresponds to summing over loops of different proper lengths. The logarithm appears because Z is defined using the multiparticle Hilbert space of QFT, serving as a generating functional for all graphs, including disconnected ones, whereas the worldline partition function $Z(t)$ is defined on the single-particle Hilbert space and corresponds to the generating functional of connected graphs, $\ln Z$ [2].

Moduli space of Torus T^2

As emphasize in the computation of the point-particle partition function, the integration measure dt/t accounts for a sum over all inequivalent loop sizes. In an analogous manner, the string partition function requires a sum over all inequivalent tori. For this purpose, we introduce the notion of the torus and discuss how to classify distinct torus geometries.

We can view the torus as the σ -plane with points identified according to

$$(\sigma_1, \sigma_2) \sim (\sigma_1, \sigma_2) + 2\pi(m, n)$$

for integers $m, n \in \mathbb{Z}$. One can show that, up to diffeomorphisms and Weyl rescalings, the most general metric on T^2 can be written in the form[3]

$$ds^2 = |d\sigma_1 + \tau d\sigma_2|^2 \quad (1.5)$$

where $\tau = \tau_1 + i\tau_2$ is a complex constant known as the *modulus* of the torus.

A suitable change of coordinates to new variables $\tilde{\sigma}^\alpha$ brings the metric locally into a unit (flat) form. Introducing the complex coordinate $w = \tilde{\sigma}_1 + i\tilde{\sigma}_2$, the torus may be equivalently described by the identification $w \sim w + 2\pi(m + n\tau)$, or in a more abstract formulation, as the quotient $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$. In these coordinates, the metric takes the simple form $ds^2 = dw d\bar{w}$, which is the flat metric on the torus subject to the above identifications[3]. In this representation, it is manifest that the modulus τ , which controls the shape of the torus, is entirely encoded in the lattice defining the identifications.

Geometrically, τ encodes the relative length of the two cycles, given by τ_2 and the skew or twist of one cycle relative to the other, given by τ_1 .

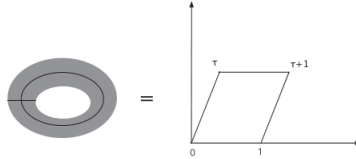


Figure 1.3: A torus may be described as a parallelogram in the plane with opposite edges identified.

Therefore, a torus may be represented as a parallelogram in the complex plane with

opposite edges identified according to $z \sim z+1 \sim z+\tau$. Because conformal transformations include local rescalings, the overall size of the parallelogram is physically irrelevant; only its shape, which is fully encoded by the parameter τ , is significant[2]. The space of conformally inequivalent tori, parametrized by τ , is called the *moduli space* $\mathcal{M}$.

From the lattice representation, one can see that the torus is invariant under the action of the modular group $PSL(2, \mathbb{Z})$, which acts on τ as[7]

$$\tau \rightarrow \frac{a\tau + b}{c\tau + d}, \quad ad - bc = 1, \quad a, b, c, d \in \mathbb{Z}, \quad (a, b, c, d) \sim (-a, -b, -c, -d) \quad (1.6)$$

This transformation changes the choice of lattice basis but does not alter the shape of the torus[2]. Therefore, the moduli space of the torus is given by the quotient $\mathcal{M} \simeq \mathbb{C}/SL(2, \mathbb{Z})$. The modular group is generated by the two basic transformations[3]

$$T : \tau \rightarrow \tau + 1, \quad S : \tau \rightarrow -\frac{1}{\tau}$$

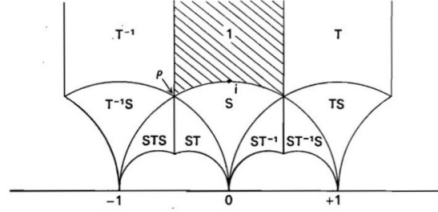


Figure 1.4: Schematic illustration of the fundamental domain of $SL(2, \mathbb{Z})$ (shaded region) in the upper half-plane, together with the actions of the modular transformations T and S that map points into the fundamental region.

Using the modular transformation $T : \tau \rightarrow \tau + 1$, one can always shift the real part of τ so that it lies within the interval[3]

$$Re \tau \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

with the endpoints of the interval identified. The transformation $S : \tau \rightarrow -1/\tau$ inverts the modulus $|\tau|$, allowing points inside the unit circle $|\tau| < 1$ to be mapped to points outside the circle $|\tau| > 1$.

By successive applications of the generators S and T , any value of τ can be mapped into the region defined by[3]

$$|\tau| \geq 1, \quad Re \tau \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

This region, depicted in Fig. 1.1, is known as the fundamental domain of $SL(2, \mathbb{Z})$.

Equivalently, the fundamental domain $\mathcal{F}$ provides a representative of the moduli space of metrics on the torus that are inequivalent under diffeomorphisms and Weyl transformations.

Also, the $SL(2, \mathbb{Z})$ -invariant measure on the fundamental domain is given by $\int \frac{d^2\tau}{(Im \tau)^2}$ [8]. To verify its invariance under $SL(2, \mathbb{Z})$ transformations, consider a general modular transformation of the form (1.6), the integration measure and the imaginary part of τ

transform as[7]

$$d^2\tau \longrightarrow \frac{d^2\tau}{|c\tau + d|^4}, \quad Im \tau \longrightarrow \frac{Im \tau}{|c\tau + d|^2}$$

As a result, the combination $\frac{d^2\tau}{(Im \tau)^2}$ remains invariant under $SL(2, \mathbb{Z})$.

Finally, notice that the globally defined translation $\sigma^a \rightarrow \sigma^a + v^a$ leaves both the metric (1.5) and the periodicity of the coordinates invariant[3]. We therefore conclude that the residual global conformal symmetry group of the torus T^2 is $U(1) \times U(1)$.

Partition function for bosonic string

Lets start by calculating the chiral partition function

$$Z(\tau) = Tr_{\mathcal{H}}(q^{L_0 - \frac{c}{24}}) \quad (1.7)$$

where $\mathcal{H}$ denotes the Fock space generated by the oscillator modes² α_m , with $m \in \mathbb{Z}$, satisfying the commutation relations $[\alpha_m, \alpha_n] = m \delta_{m+n, 0}$. The ground state $|0\rangle$ of the Hilbert space is defined by

$$\alpha_m |0\rangle = 0, \text{ for } m \geq 0, \quad \langle 0|0\rangle = 1$$

An orthonormal basis of $\mathcal{H}$ is given by

$$|\{n_k\}\rangle \equiv |n_1, n_2, \dots\rangle = \mathcal{N}_{\{n_k\}} \alpha_{-1}^{n_1} \alpha_{-2}^{n_2} \dots |0\rangle, \quad n_k \geq 0$$

with inner products

$$\langle n_1, n_2, \dots | n'_1, n'_2, \dots \rangle = \delta_{n_1, n'_1} \delta_{n_2, n'_2} \dots$$

For later reference, we recall the expression of the Virasoro zero mode L_0 in terms of the oscillators:

$$L_0 = \frac{1}{2} \alpha_0^2 + \sum_{n=1}^{\infty} \alpha_{-n} \alpha_n$$

which satisfies

$$[L_0, \alpha_m] = -m \alpha_m, \quad L_0 |n_1, n_2, \dots\rangle = \sum_{k=1}^{\infty} k n_k |n_1, n_2, \dots\rangle$$

We now compute the partition function by restricting attention to the contributions from the oscillator modes³. Evaluating the trace in (1.7) in the occupation-number basis yields[2]

²Recall that each scalar field X decomposes into a zero mode together with an infinite tower of harmonic oscillator modes α_m , which create states of energy m .

³This means that, at this stage, we are not including the contribution arising from the zero mode a_0 .

$$\begin{aligned}
Z_{osc}(\tau) &= Tr_{\mathcal{H}_{osc}} \left(q^{L_0 - \frac{1}{24}} \right) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \langle n_1, n_2, \dots | q^{L_0 - \frac{1}{24}} | n_1, n_2, \dots \rangle \\
&= q^{-\frac{1}{24}} \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots q^{\sum_{k=1}^{\infty} k n_k} = q^{-\frac{1}{24}} \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \prod_{k=1}^{\infty} q^{k n_k} \\
&= q^{-\frac{1}{24}} \prod_{k=1}^{\infty} \sum_{n_k=0}^{\infty} q^{k n_k} = q^{-\frac{1}{24}} \prod_{k=1}^{\infty} \frac{1}{1 - q^k}
\end{aligned}$$

where we have set $c = 1$ and performed a geometric series summation in the final step.

The result obtained above can be conveniently expressed in terms of the Dedekind eta function. In particular, the contribution of the oscillator modes of a single scalar field is[2]

$$Z_{osc}(\tau) = \eta^{-1}(q), \quad q = e^{2\pi i \tau}$$

An analogous contribution arises from the antiholomorphic sector involving $\bar{q}$ and $\bar{L}_0$. Including both the holomorphic and antiholomorphic parts, the full oscillator partition function becomes[2]

$$Z_{osc}(\tau, \bar{\tau}) = Tr_{\mathcal{H}_{osc}} \left(q^{L_0 - \frac{1}{24}} \bar{q}^{\bar{L}_0 - \frac{1}{24}} \right) = |\eta(\tau)|^{-2}$$

where the trace taken over oscillator modes only.

This result generalizes straightforwardly to d independent oscillator modes. Each degree of freedom contributes a factor of $|\eta(\tau)|^{-2}$, accounting for both left- and right-moving excitations. Consequently, the full oscillator partition function is

$$Z_{osc}^{(d)}(\tau, \bar{\tau}) = |\eta(\tau)|^{-2d}$$

At this stage, the remaining contribution to the partition function arises from the zero mode p of the scalar field. The zero mode of the scalar field is encoded in the operator a_0 , and the Virasoro zero mode can be written as $L_0 = \frac{\alpha'}{4} p^2 + N$ where N denotes the oscillator number operator. A complete basis of states is therefore given by $|p; \{n_k\}\rangle$, labelled by the continuous momentum p and the occupation numbers $\{n_k\}$ of the oscillator modes. The contribution of the zero mode is[7]

$$(q\bar{q})^{\frac{\alpha'}{4} p^2} |p; \{n_k\}\rangle = \exp(-\pi \alpha' \tau_2 p^2) |p; \{n_k\}\rangle \quad (1.8)$$

where we have used $q = e^{2\pi i \tau}$ and $\bar{q} = e^{-2\pi i \bar{\tau}}$. The trace appearing in the partition function requires a sum over all states, which factorizes into a trace over the oscillator modes and an integral over the continuous momentum eigenvalues⁴:

$$\begin{aligned}
Tr \left((q\bar{q})^{\frac{\alpha'}{4} p^2} q^N \bar{q}^{\bar{N}} \right) &= \int \frac{dp}{2\pi} \sum_{\{n_k\}, \{\bar{n}_k\}} \langle p; \{n_k\}, \{\bar{n}_k\} | (q\bar{q})^{\frac{\alpha'}{4} p^2} q^N \bar{q}^{\bar{N}} | p; \{n_k\}, \{\bar{n}_k\} \rangle \\
&= Tr_{\mathcal{H}_{zero \text{ mode}}} \left((q\bar{q})^{\frac{\alpha'}{4} p^2} \right) Z_{osc}(\tau, \bar{\tau})
\end{aligned} \quad (1.9)$$

⁴The factor $\frac{dp}{2\pi}$ appears because the zero-mode momentum p has a continuous spectrum, and the trace over its eigenstates is replaced by an integral normalized per unit 2π .

where the trace over the zero mode is[7]:

$$Z_{\mathcal{H}_{\text{zero mode}}} = \text{Tr} \left((q\bar{q})^{\frac{\alpha'}{4} p^2} \right) = V_1 \int \frac{dp}{2\pi} e^{-\pi \alpha' \tau_2 p^2} = V_1 (4\pi^2 \alpha' \tau_2)^{-\frac{1}{2}}$$

where the inner product over the momentum states gives a factor of the spacetime volume $\langle p|p \rangle = \delta^{(1)}(p-p) \equiv V_1$

Having d independent modes, which correspond to a d -dimensional spacetime, the partition function of the zero modes is given by

$$Z_{\mathcal{H}_{\text{zero mode}}}^{(d)} = V_d (4\pi^2 \alpha' \tau_2)^{-\frac{d}{2}} \quad (1.10)$$

where V_d is the spacetime volume.

To construct the string partition function rigorously, one should work in covariant quantization and include the ghost fields; we will comment on this aspect later. For simplicity, however, we will present the computation in the lightcone gauge.

In lightcone gauge, there are 24 transverse oscillator modes, while 26 zero modes exist [8]. Combining these contributions, the worldsheet partition function reads[7]

$$Z_{\text{LC}}(\tau, \bar{\tau}) = V_d (4\pi^2 \alpha' \tau_2)^{-13} |\eta(q)|^{-48}$$

where V_d denotes the spacetime volume. To obtain the spacetime partition function, one must divide by the volume of the conformal Killing group, which is simply $U(1) \times U(1)$, representing translations along the torus cycles[7]. The corresponding volume is $\text{Vol} = 4\pi^2 \text{Im } \tau$. Next, we integrate over the moduli space of the torus, i.e., over all inequivalent tori. Neglecting constant prefactors, the closed string partition function becomes[8]

$$Z_{\text{closed string}} = \int_{\mathcal{F}} \frac{d^2 \tau}{\tau_2} (\alpha' \tau_2)^{-13} |\eta(q)|^{-48} \quad (1.11)$$

where $\mathcal{F}$ denotes the fundamental domain of the modular group.

In lightcone gauge, we need to impose the level-matching condition, which we have not yet implemented in our computations[3]. We will now explain how the level-matching condition arises from the integration over the fundamental domain.

The free string describes an infinite tower of particle states with masses

$$m_n^2 = \frac{4}{\alpha'} (n-1), \quad n = 0, 1, 2, \dots$$

We will see that a natural interpretation of equation (1.11) is that the string partition function can be expressed as a sum over vacuum loops of these particles[8].

Equation (1.4) provides the partition function for a single particle of mass m . For string theory in $D = 26$ dimensions, this leads to⁵

$$Z = \int_0^\infty dt \frac{1}{t^{14}} \sum_{n=0}^\infty e^{-m_n^2 t}$$

⁵This is the one-loop partition function of the bosonic string. No logarithm appears because the amplitude is written directly in the first-quantized formalism; nevertheless, it is equivalent to $\ln Z$ in the second-quantized formulation of the point particle.

The mass spectrum can also be written in terms of the L_0 and $\tilde{L}_0$ operators:

$$m^2 = \frac{4}{\alpha'}(L_0 - 1) = \frac{4}{\alpha'}(\tilde{L}_0 - 1) = \frac{2}{\alpha'}(L_0 + \tilde{L}_0 - 2)$$

subject to the level-matching constraint $L_0 = \tilde{L}_0$. Level matching can be imposed using the integral representation of the Kronecker delta[3]:

$$\delta_{L_0, \tilde{L}_0} = \frac{1}{2\pi} \int_{-1/2}^{+1/2} ds e^{2\pi i s(L_0 - \tilde{L}_0)}$$

Replacing the sum over particle species with a trace over the string spectrum subject to level matching, the partition function becomes[8]

$$Z = \int_0^\infty dt \frac{1}{t^{14}} \int_{-1/2}^{+1/2} ds \text{Tr} \left[e^{2\pi i s(L_0 - \tilde{L}_0)} e^{-\frac{2t}{\alpha'}(L_0 + \tilde{L}_0 - 2)} \right]$$

We can introduce a complex parameter τ , which combines the length of the loop t and the auxiliary variable s used to impose level matching: $\tau = s + \frac{2it}{\alpha'}$ [8]. Performing the trace over the string spectrum yields the Dedekind eta functions, just as before. Using the standard definition $q = e^{2\pi i \tau}$, we then obtain:

$$Z_{\text{closed string}} = \int_R \frac{d^2 \tau}{\tau_2} (\alpha' \tau_2)^{-13} |\eta(q)|^{-48} \quad (1.12)$$

Here the region of integration R is

$$R: \quad \tau_2 > 0, \quad |\tau_1| < \frac{1}{2}$$

The amplitude 1.11 for a single point particle diverges as $t \rightarrow 0$, which is the familiar UV divergence in quantum field theory[3]. Summing over the entire string spectrum only exacerbates this divergence, since all string states contribute with the same sign. Nevertheless, if we compare the sum 1.12 with the full string amplitude 1.11, we observe a strong resemblance, along with one important distinction: the integrands are the same, but the domains of integration differ[3]. In string theory, the integration is restricted to the fundamental region $\mathcal{F}$ while in field theory the integration extends over the larger region R .

We will simply state how the partition function is obtained using the bc ghost system. Gauge fixing requires the introduction of the bc ghosts, and in critical bosonic string theory the spacetime dimension is $D = 26$ [3]. The ghost contribution precisely cancels the contribution from the two non-transverse (longitudinal and timelike) polarizations[7].

$$|\eta(\tau)|^{-48} = \left| \underbrace{\eta(\tau)^{-26}}_{X^\mu} \cdot \underbrace{\eta(\tau)^2}_{bc} \right|^2$$

This provides a clear demonstration of how the ghost sector restores unitarity by canceling the non-physical excitations. The underlying reason for this cancellation lies in the anticommuting nature of the ghost field[7]. After including the ghost contribution, we recover the same expression as in equation (1.11).

Modular invariance and UV finite

A crucial property of the partition function Z is its modular invariance, namely invariance under $\text{PSL}(2, \mathbb{Z})$ transformations of the torus[8].

The origin of this invariance lies in the transformation properties of the Dedekind eta function, which is an example of a modular form, i.e. a function of τ with well-defined behavior under $\text{PSL}(2, \mathbb{Z})$ transformations.

The modular group is generated by the S and T transformations, under which the Dedekind eta function transforms as[2]

$$\eta\left(-\frac{1}{\tau}\right) = (-i\tau)^{\frac{1}{2}}\eta(\tau), \quad \eta(\tau+1) = e^{\frac{i\pi}{12}}\eta(\tau)$$

Modular invariance of the torus partition function Z_{T^2} follows from the fact that the measure $\frac{d^2\tau}{\tau_2^2}$ and the combination $\tau_2|\eta(\tau)|^4$ are each individually modular invariant[8].

The limit $\tau_2 \rightarrow 0$ corresponds to the ultraviolet (UV) regime, since the Euclidean time direction becomes very short, representing high-energy processes[7]. In this limit, the integral 1.11 would naively diverge. Geometrically, this corresponds to particles propagating around arbitrarily small loops[8].

In string theory, however, this UV-divergent region is excluded due to modular invariance of the torus, while it remains present in point-particle field theory[8]. We therefore arrive at the important conclusion that the one-loop string partition function Z is UV finite. Indeed, modular invariance acts as an intrinsic UV cutoff and removes the divergences that appear in analogous point-particle theories[7]. This mechanism underlies the ultraviolet finiteness of the one-loop string amplitude.

Although demonstrated here for the simplest vacuum amplitude, the same mechanism continues to operate for higher n -point functions[7].

The infrared (IR) regime corresponds to the limit $\tau_2 \rightarrow \infty$, in which the torus (or equivalently a cylinder with identified ends) becomes very long[8]. In this limit, we may expand the eta function as[7]

$$\int_{-\infty}^{\infty} d^2\tau |\eta(\tau)|^{-48} = \int_{-1/2}^{1/2} d\tau_1 \int_{-\infty}^{\infty} d\tau_2 (q\bar{q})^{-1} (1+24q+\dots)(1+24\bar{q}+\dots) = \int_{-\infty}^{\infty} d\tau_2 ((q\bar{q})^{-1} + 24^2 + \dots)$$

where terms linear in q or $\bar{q}$ vanish upon integration over τ_1 . From this we deduce the infrared behavior[7]

$$iV_{26} \int_{-\infty}^{\infty} \frac{d\tau_2}{2\tau_2} (4\pi^2 \alpha' \tau_2)^{-13} \left(\underbrace{e^{4\pi\tau_2}}_{\text{tachyon}} + \underbrace{24^2}_{\text{Massless modes}} + \underbrace{\dots}_{\text{Massive modes}} \right)$$

The first term is divergent, but this divergence is an artifact of the presence of the tachyon[8]. Since the tachyon is removed in superstring theory, this divergence can be safely ignored. The next contribution arises from massless states. Consequently, the long-distance behavior of the theory is governed by massless modes, whose contribution to the partition function is finite[7].

BIBLIOGRAPHY

- [1] M. L. Bellac, *Thermal Field Theory*, ser. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 3 2011.
- [2] T. Mohaupt, *A Short Introduction to String Theory*. Cambridge University Press, 3 2022.
- [3] J. Polchinski, *String Theory*, ser. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 1998.
- [4] R. Blumenhagen, D. Lüst, and S. Theisen, *Basic concepts of string theory*, ser. Theoretical and Mathematical Physics. Heidelberg, Germany: Springer, 2013.
- [5] M. D. Schwartz, *Quantum Field Theory and the Standard Model*. Cambridge University Press, 3 2014.
- [6] A. M. Polyakov, *Gauge Fields and Strings*. London: Taylor & Francis, 1987.
- [7] T. Weigand, “Introduction to string theory (lecture notes),” Lecture notes, Univ. Heidelberg, Winter Term 2011/12, 2011.
- [8] D. Tong, “String theory: Lecture notes,” <https://www.damtp.cam.ac.uk/user/tong/string/string.pdf>, 2009, part III lecture notes; freely available online.